# Midterm Delivery

CSC1180

COMBUS4

## BumpBudget

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# 1. Introduction

## 1.1 Overview

BumpBudget is an integrated financial planning platform designed specifically for expecting and new parents. The system provides a combination of practical budgeting tools, maternity leave planning, savings goal tracking, and educational resources related to pregnancy and childcare.

The platform is intended for individuals or couples who are planning for a child, currently expecting, or have recently become parents. Its main goal is to make financial planning during this challenging stage simpler, more accurate, and personalised. By linking a user's due date, income, and financial goals with predictive budgeting features, BumpBudget helps users anticipate upcoming costs, manage reduced income during maternity or paternity leave, and make informed financial decisions. The core idea is to transform financial uncertainty into proactive planning, reducing stress and supporting confident decision-making.

## 1.2 Business Context

### 1.2.1. Current Market Situation

In the UK and Ireland, there are several apps targeting parents and pregnancy-related needs:

1. Pregnancy/maternity apps like Emma's Diary focus on tracking milestones, appointments, and general pregnancy content.
2. Financial planning resources such as MoneyHelper (UK) and Baby Steps (Ireland) provide timelines for baby-related costs and benefit/allowance information.
3. General budgeting apps like Snoop or Money Dashboard offer broad financial management but do not address pregnancy-specific challenges.

Despite this, no single platform integrates financial planning with pregnancy milestones, maternity leave planning, and predictive cost forecasting. Current solutions either focus on general budgeting or pregnancy tracking, leaving parents to navigate multiple tools for a full picture of their financial situation.

### 1.2.2. Need and Market Gap

Expecting and new parents face complex financial challenges, including: predicting medical costs and baby-related expenses, adjusting budgets for reduced income during maternity/paternity leave, understanding government benefits and allowances and planning for childcare and education expenses.

There is a clear gap in the market for a pregnancy-aware financial planner that combines budgeting, predictive modelling, and educational support into a single platform. BumpBudget addresses this by: forecasting costs tied to pregnancy stages and due dates, providing clear visibility of financial obligations before and after birth, and offering educational content that integrates with financial planning.

Research has shown that major life transitions such as pregnancy and early parenthood are often associated with increased financial stress, uncertainty, and reduced wellbeing, particularly when combined with income disruption and rising household costs (McCarthy and Collins, 2023).

### 1.2.3. Long-Term Considerations

BumpBudget could adopt monetisation strategies such as: freemium access with premium features like advanced forecasting and personalised consultations, partnerships with financial institutions or baby product companies and subscription-based access for enhanced planning tools and content.

By filling a unique niche at the intersection of financial planning and pregnancy support, BumpBudget positions itself as an essential tool for parents navigating one of life's most financially significant periods.

## 1.3 Glossary

| | |
|---|---|
| **Chart.js** | A JavaScript library used to display interactive financial charts and visualisations on the user dashboard. |
| **Django** | A Python-based web framework used to implement the backend logic, request handling, and security of the system. |
| **Django ORM** | A tool within Django that enables database interaction using Python objects instead of raw SQL queries. |
| **Forecasting Engine** | A system component that generates predictions of future income, expenses, and savings using user data and time-based inputs. |
| **GDPR** | A data protection regulation that governs how personal and financial data must be stored and processed securely. |
| **MySQL** | A relational database management system used to store user, budget, expense, and milestone data. |
| **Three-Tier Architecture** | A system design pattern consisting of frontend, backend, and database layers. |

## 1.4 Initial Business Model Summary

### 1.4.1. Project Concept

BumpBudget is a digital financial planning platform designed specifically for expecting and new parents in Ireland. The project addresses a clear gap in the market: while budgeting tools and pregnancy apps exist, none integrate pregnancy timelines, maternity leave entitlements, and baby-related costs into a single, structured financial planning experience. BumpBudget supports users through pregnancy and early parenthood by transforming fragmented financial information into a clear, step-by-step plan tailored to their life stage.

### 1.4.2. Value Proposition

BumpBudget's value proposition centres on clarity, confidence, and reduced financial stress during pregnancy and early parenthood. Based on primary and secondary research, parents frequently feel overwhelmed by reduced income during maternity leave, underestimated baby costs, and unclear entitlements. BumpBudget addresses these pain points by offering:

- Personalised budgeting and income forecasting linked to pregnancy milestones.
  Clear comparisons of public, semi-private, and private maternity care costs.
- Tools for planning savings and anticipating major expenses (e.g. childcare deposits).
- Reminders for key benefit deadlines and appointments.
- Trusted, Ireland-specific information delivered in one central platform.

The platform delivers both functional value (budgeting, forecasting) and emotional value (peace of mind and reassurance), which is a key driver of adoption in family-related digital tools.

#### 1.4.3. Target Users and Customers

The primary customer segments identified through research are:

- Expecting mothers in Ireland, particularly first-time parents who require guidance on maternity leave entitlements and upcoming costs.
- New parents with children aged 0–2, who need support managing reduced income alongside increasing childcare and household expenses.

A secondary customer segment includes employers and insurers, who benefit from offering structured financial planning tools to employees during parental leave, reducing HR workload and improving employee wellbeing.

#### 1.4.4. Market Size

BumpBudget operates within the growing personal finance and family-support app market. While the overall personal finance app market is large, BumpBudget focuses on a narrower, well-defined segment: expecting and new parents in Ireland. Research indicates a strong demand for pregnancy-related financial guidance, particularly given rising childcare and cost-of-living pressures. By targeting users at a specific and financially sensitive life stage, BumpBudget positions itself within a clearly identifiable and reachable niche rather than a saturated general budgeting market.

#### 1.4.5. Revenue Streams

BumpBudget adopts a freemium business model to ensure accessibility while supporting long-term sustainability. Proposed revenue streams include:

- Premium subscriptions for advanced budgeting, forecasting, and scenario planning tools.
- Affiliate partnerships with baby retailers, childcare providers, and financial institutions.
- B2B partnerships with employers and HR departments offering BumpBudget as part of employee wellbeing programmes.
- Anonymised, aggregated insights (handled under strict GDPR compliance) to inform maternity-related policy and market trends.

These revenue streams align with user expectations and avoid placing financial barriers on parents during an already costly life transition.

# 2. General Description

## 2.1. Product/System Function

BumpBudget provides a suite of financial planning tools for expecting and new parents, combining budgeting, forecasting, and educational support in a single platform. The system functions can be categorised into user functionality and system functionality.

#### 2.1.1. User Functionality:

- **Register/Login**: Users can create an account and log in securely.
- **Create Budget**: Users can set up a personalised budget based on income, maternity/paternity leave, and expected expenses.
- **Update Expenses**: Users can log and track daily and monthly expenses related to pregnancy, baby essentials, and household costs.
- **View Forecasts**: Users can see predictive visualisations of expenses and savings over time, including projections for maternity leave and baby-related costs.
- **Track Milestones**: Users can monitor pregnancy stages and newborn milestones, with financial implications for each stage highlighted.

#### 2.1.2. System Functionality:

- **Forecasting Engine**: Uses historical data, predictive algorithms, and time-series analysis to generate expense and savings forecasts.
- **Database Handling**: Securely stores user information, budgets, expenses, and milestone data using MySQL.
- **Dashboard Generation**: Presents an intuitive, interactive dashboard displaying budgets, forecasts, and milestone-related costs.
- **Educational Content Delivery**: Provides articles, checklists, and tips on benefits, allowances, and cost-saving strategies.

## 2.2 User Characteristics and Objectives

#### 2.2.1. Target Users:

- Expecting parents, first-time mothers and fathers.
- Parents with young children under 3 years old.
- Low- to middle-income households seeking structured financial planning.

#### 2.2.2. User Abilities:

- Basic digital literacy, comfortable with mobile apps or web platforms.
- Familiarity with common budgeting concepts.

#### 2.2.3. User Goals:

- Reduce financial stress during pregnancy and early parenthood.
- Plan for maternity/paternity leave income changes.
- Understand and anticipate pregnancy- and child-related costs.
- Track savings goals and manage everyday household expenses.

## 2.3 Operational Scenarios

#### 2.3.1. Registering and Logging In:

- The user visits the BumpBudget site and selects “Register”.
- Provides basic information (name, email, password).
- Receives confirmation email and logs in.
- Accesses the personalised dashboard.

**2.3.2. Creating a Pregnancy-Linked Budget:**

- User inputs income, expected maternity/paternity leave dates, and known expenses.
- System prompts for pregnancy stage or due date.
- The budget is created with suggested allocations for essentials, healthcare, and savings.

**2.3.3. Updating Expenses:**

- User logs daily or monthly expenses.
- The system categorises entries (e.g., baby essentials, household, healthcare).
- Dashboard and forecasts update automatically.

**2.3.4. Viewing Financial Forecasts:**

- The user accesses the forecast panel.
- Visualisations show expected income, predicted expenses, and projected savings.
- Milestone events such as medical appointments or baby's first month costs are highlighted.

**2.3.5. Checking Milestone-Related Costs:**

- User selects a milestone (e.g., second trimester, baby under 6 months).
- The system displays estimated costs associated with that stage.
- Users can adjust budget or savings goals accordingly.

## 2.4 Constraints

**Time Constraints:**

- Development must be completed within the academic/project timeline.
- Limited time for testing predictive algorithms extensively.

**Data Limitations:**

- Availability and accuracy of financial and pregnancy-related data may vary.
- Reliance on user-provided inputs can introduce inconsistencies.

**Security Requirements:**

- Sensitive financial and personal data must be encrypted and GDPR-compliant.
- Authentication and access control need to be robust.

**Technical Constraints:**

- Forecasting engine complexity is limited by computational resources.
- The system must remain lightweight and responsive for web access.

**Legal Constraints:**

- GDPR and other data protection regulations must be adhered to.
- Any financial advice must be presented as guidance rather than professional consultation

# 3. Functional Requirements

## 3.1 Register

The registration process allows new users to create an account and access personalised budgeting tools.

| Required Data Fields | Importance | Technical Implementation | Dependencies |
|---|---|---|---|
| Full name, email address, and a password that meets minimum security requirements. | Enables a personalised user experience while ensuring secure access and data protection. | Handled using Django's built-in authentication system. | The login functionality depends on successful registration, and a corresponding user record is created in the database to store all related financial and milestone data. |
| Optionally, users may also enter their due date and expected number of children, which allows the system to tailor budgets, forecasts, and milestone tracking. | It also links each user to their individual budgets, expenses, and pregnancy milestones stored within the system. | Passwords are securely hashed using Django's default security methods. | |
| | | Email verification is optional but recommended to improve account validity. | |

### 3.2 Log In

The login feature allows registered users to securely access their BumpBudget account.

| Functionality | Password Recovery | Security Considerations | Dependencies |
|---|---|---|---|
| Users log in by entering their email address and password. An optional | If a user forgets their password, they can reset it through a "Forgot Password" option, which sends a secure password reset link to their registered email address. | The system enforces password complexity rules, limits repeated failed login attempts through account lockout mechanisms, and ensures all communication is encrypted using HTTPS. | This functionality requires prior registration and access to the database for authentication and secure session management. |
| "Remember Me" feature allows users to remain logged in across sessions. | | | |

### 3.3 Create Budget Plan

The budget creation feature allows users to generate a personalised financial plan based on their pregnancy and financial situation.

| Inputs | Process | Dependencies |
|---|---|---|
| Current income. | Tailored budget template. | Stores all user inputs and generated budgets. |
| Expected income during maternity or paternity leave. | | |
| Pregnancy stage or due date. | Calculates initial forecasts for future expenses and savings. | The forecasting engine uses tools such as Pandas and Prophet to produce financial predictions. |
| Planned expenses including baby essentials, healthcare, and childcare. | | |

### 3.4 Update Budget / Expenses

This feature allows users to keep their financial plan up to date as their circumstances change.

| Functionality | System Response | Dependencies |
|---|---|---|
| Users can add, edit, or remove expenses. | Whenever updates are made, the system automatically recalculates forecasts and refreshes all dashboard visualisations. | All changes are securely stored in the database. |
| Update maternity or paternity leave details. | | |
| Adjust savings goals or financial priorities. | Historical records are preserved to allow users to review past spending patterns. | The forecasting engine recalculates predictions each time new data is entered. |

### 3.5 View Financial Dashboard

The financial dashboard is the main interface users interact with when using BumpBudget.

| Overview | Visualisation | Technical Implementation |
|---|---|---|
| The dashboard presents a clear summary of income, spending, savings, and upcoming costs in one place. | Interactive charts display spending patterns and financial progress over time. | User data is retrieved from the database using Django. |
| Allowing users to quickly understand their financial position. | Upcoming expenses, such as medical appointments or baby-related purchases, are highlighted to help users plan ahead. | Data is then displayed through interactive charts created with Chart.js, ensuring the information is accurate, current, and easy to interpret. |

### 3.6 Monitor Pregnancy Milestones / Forecasts

BumpBudget links pregnancy milestones with financial planning to help users understand how different stages affect their finances.

| Milestone Tracking | Forecasting and Visualisation |
|---|---|
| After a user enters their due date, the system tracks key stages such as each trimester, birth, and early parenthood. | Financial forecasts update automatically as milestones are reached, reflecting changes in income and spending. |
| Each stage is linked to typical costs, allowing users to anticipate upcoming expenses. | These forecasts are displayed through simple charts on the dashboard, helping users assess their financial readiness and adjust budgets or savings goals in advance. |

### 3.7 Database

The database is a core component of the BumpBudget system, as it stores and manages all user and financial data required for the platform to function effectively.

| Stored Data | Purpose and Importance | Data Handling and Queries |
|---|---|---|
| The database stores user account information, including:<br>- authentication details<br>- financial data such as budgets, income entries, expenses, pregnancy milestones, and generated financial forecasts. | A centralised database is essential for maintaining data consistency, enabling personalised user experiences, and ensuring that financial forecasts and dashboards remain accurate and up to date. | BumpBudget uses MySQL as a relational database, with Django's Object Relational Mapping (ORM) managing database queries. This allows secure, efficient data retrieval and updates without requiring direct SQL queries. |
| This data allows the system to personalise budgeting tools and link financial information to specific pregnancy stages. | It allows users to return to their accounts and view historical data, track progress over time, and make informed financial decisions. | All data access is controlled through the backend to ensure data integrity, performance, and compliance with data protection requirements. |

# 4. System Architecture

### 4.1 Architectural Diagram for the System (*Fig 1*)

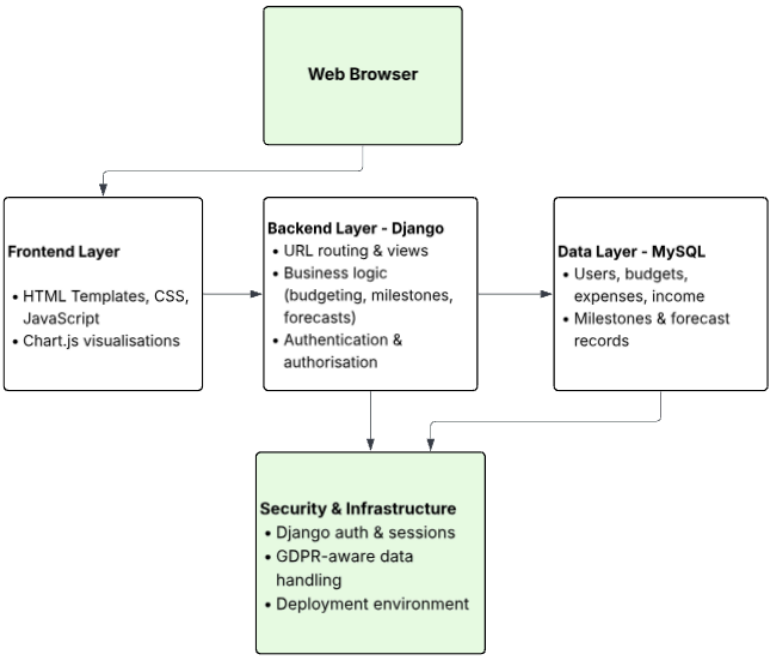



*Figure 1 presents the high-level three-tier architecture of the BumpBudget system, showing the separation between the frontend, Django backend, and MySQL data layer.Cross-cutting security and infrastructure components are included to indicate system-wide concerns such as authentication and data protection.*

BumpBudget uses a three-tier architecture where users interact through a web browser, which connects to the Django backend to handle business logic, authentication, and user actions. The backend communicates with a MySQL database to securely store budgets, expenses, income and milestone data. Security and infrastructure support all layers, ensuring GDPR-compliant data handling, protected sessions, and a stable deployment environment.

### 4.2 Web Application

BumpBudget's web application is designed as a timeline-aware budgeting interface, meaning the user experience changes depending on pregnancy stage and upcoming milestones. The frontend is built with Django templates using a shared base layout for consistent navigation and structure, while individual pages extend this layout for specific features such as registration, login, dashboard views, budgeting inputs, and milestone-related cost displays. This template-driven structure keeps the interface consistent and scalable as new features are added (e.g., milestone tracker, leave calculator, or savings goals).

The dashboard is designed to reduce cognitive load by presenting users with "at-a-glance" financial clarity. For example, it highlights key metrics such as monthly income, expenses, remaining budget, and upcoming costs tied to pregnancy or newborn stages, allowing users to immediately understand where they stand financially without having to interpret raw data tables. Interactive visualisations (Chart.js) translate financial data into clear charts (e.g., spending breakdowns and trend forecasting), which is especially important for users who may feel overwhelmed by financial planning during pregnancy. Prior research indicates that well-designed financial dashboards which reduce cognitive load can significantly improve user understanding and decision-making, particularly in data-intensive contexts (Cheng et al., 2024).

The web application also supports responsiveness and accessibility so it can be used easily across laptop and mobile screens, which fits real parent use-cases (on-the-go, quick check-ins, small bursts of time). This makes the product more realistic than a static prototype because it demonstrates an implementable, user-centred interface rather than "screens only."

## 4.3 Django Backend

Django is used as the main backend framework for the BumpBudget system. It handles the core logic of the application, including user actions such as registering, logging in, creating a budget, adding expenses, and viewing dashboard information. When a user performs an action, the request is passed through Django's URL routing system to the appropriate backend view, where the data is checked, processed, and stored in the database before the updated information is displayed back to the user on the website. This structure helps keep the frontend and backend separate, which makes the system easier to manage and build on as the project develops.

One of the main reasons Django was chosen is its built-in security features, which are important because the system handles personal and financial data such as income, expenses, and due dates. Django includes secure user authentication, password hashing, and session management by default, which reduces the risk of security issues compared to building these features from scratch. It also provides protection against common web vulnerabilities such as CSRF attacks and allows access controls to be put in place so users can only view and manage their own data. Using these built-in features helps ensure the system is secure while remaining suitable for an academic project.

Given the sensitivity of personal and financial data, compliance with data protection regulations such as GDPR is essential, and prior studies emphasise the importance of secure authentication, controlled data access, and privacy-aware system design in web-based financial applications (OECD, 2024).

Django's ORM also strengthens the implementation by allowing the system to interact with MySQL using model objects instead of raw SQL. This improves developer productivity, reduces risk of injection vulnerabilities, and supports structured relationships (e.g., one user → many expenses, one budget → many categories, one profile → milestones and predictions). It also supports future features like audit logs, trend tracking, and personalised forecasting pipelines without needing to redesign the database layer.

A relational MySQL database supports BumpBudget's backend, enabling structured, secure storage of user financial data over time. Through Django's ORM, the system models relationships between users, budgets, expenses, and pregnancy milestones, which is essential for both personalised dashboards and predictive analysis. This historical, time-stamped data forms the foundation for the machine learning forecasting module, allowing the system to analyse spending trends, identify risk periods, and generate forward-looking recommendations. Using MySQL also supports scalability, ensuring the application can handle increasing data volumes as more users and longer financial histories are introduced.

## 4.4 Machine Learning / Forecasting

Time-series forecasting methods that combine statistical and machine learning approaches have been extensively explored in the literature, demonstrating strong potential for improving predictive accuracy when modelling future financial trends (Makridakis et al., 2023). Building on this research, BumpBudget applies forecasting techniques within a personalised financial planning context to anticipate future expense patterns and income changes over time.

What distinguishes BumpBudget from generic budgeting tools is that its forecasting is pregnancy-linked and time-aware. Rather than simply tracking historical spending, the platform incorporates a dedicated forecasting module that predicts upcoming expense patterns based on (1) user-provided inputs such as income, expected leave dates, due date, and current spending behaviour, (2) stage-based milestone costs including trimester-specific needs, hospital expenses, and newborn essentials, and (3) accumulated historical expense data. This enables a proactive system that can identify and warn users about upcoming "budget pressure points", such as income reductions during maternity leave, childcare deposits, or high-cost purchase periods.

Recent research highlights that machine learning has been widely applied in financial contexts such as risk assessment, portfolio optimisation, and predictive modelling, demonstrating its effectiveness for automated decision support systems (Nazareth & Reddy, 2023; Gao et al., 2024). In line with this research, the machine learning component of BumpBudget is designed to support three key intelligent outputs that enhance decision-making and user awareness.

First, expense forecasting uses time-series prediction to estimate likely spending totals in future weeks or months, which are visualised on the dashboard to support forward planning rather than reactive budgeting. Second, financial risk classification evaluates whether a user is trending toward overspending or potential shortfalls (e.g., "on track", "at risk", or "over budget"), allowing the system to provide early warnings and targeted insights. Third, budget recommendations generate suggested adjustments to spending allocations, such as increasing savings ahead of reduced income periods or reallocating funds as milestone-related costs approach.

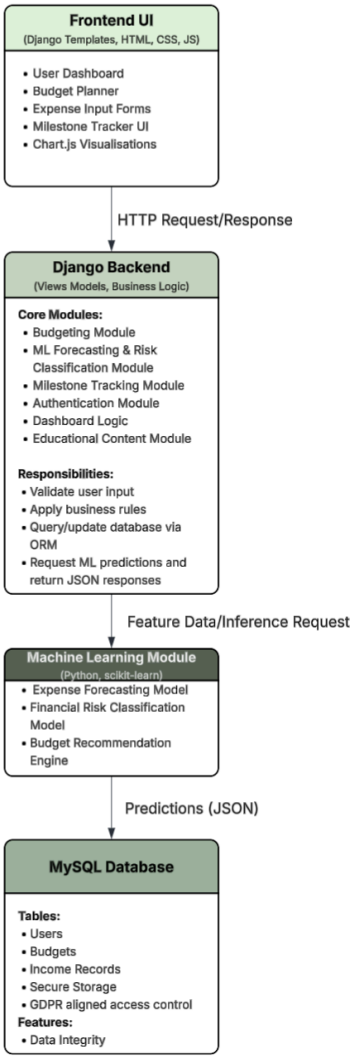
This approach differentiates BumpBudget from typical budgeting applications by combining life-stage awareness with predictive modelling. While many existing tools focus on either pregnancy tracking or generic budgeting, few integrate predictive, pregnancy-aware financial forecasting within a single platform. As a result, BumpBudget supports proactive financial planning during a clearly defined and financially sensitive life transition.

§To ensure feasibility within project constraints, the forecasting module is designed using a staged development approach. Initial implementations rely on rule-based or experience based predictions derived from milestone timing and observed expense patterns, while the system architecture supports progression toward more advanced machine learning techniques as sufficient user data becomes available. This balances innovation with practical engineering considerations and demonstrates a scalable, research-informed system design (Makridakis et al., 2023).

# 5. High-Level Overview

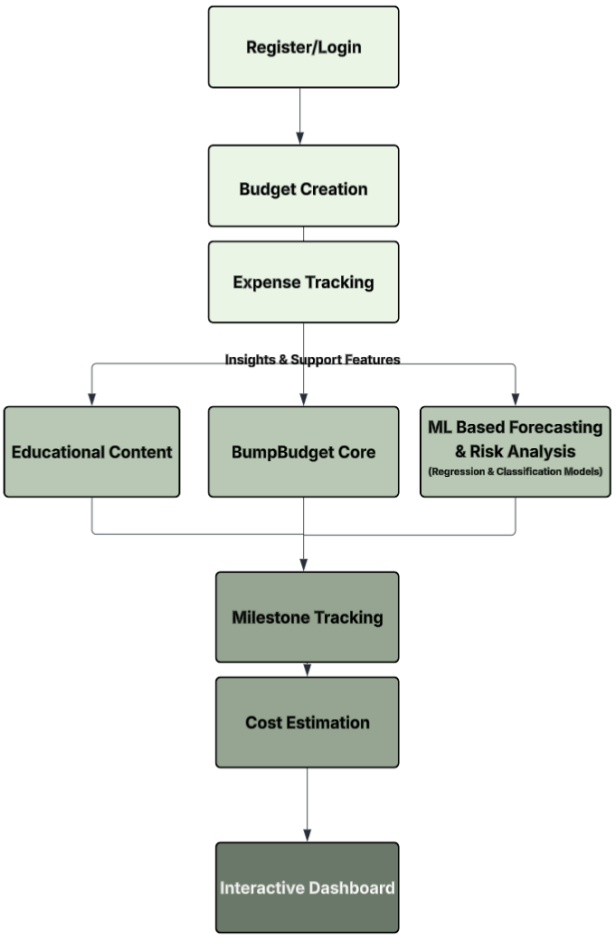
## 5.1 High Level Design Diagram

(*Fig 2*)




## 5.2 High Level Feature Overview

(*Fig 3*)




*(Fig 2) - This diagram illustrates the overall system architecture, showing the interaction between the frontend interface, Django backend, MySQL database, and a dedicated machine learning module used for expense forecasting, financial risk classification, and budget recommendations.*

*(Fig 3)- This diagram presents the user-facing feature flow of the system, highlighting how expense tracking data is processed through ML-based forecasting and risk analysis to generate cost estimations and insights within the interactive dashboard.*

# 6. Preliminary Schedule

## 6.1 Project Timeline

The BumpBudget project is organised into a series of phases running from initial research through to final delivery. The early phase focuses on problem definition, market analysis, and requirements specification. This is followed by system design and architecture planning, before moving into backend and frontend development using Django and MySQL. The later stages of the project concentrate on system integration, testing, refinement, and final documentation. Figure 6.3 presents the Gantt chart outlining the timeline, task sequencing, and allocation of responsibilities across the project duration.

## 6.2 Task List

The development of BumpBudget has been planned from the beginning of the semester and will continue through to the final delivery. In the early weeks of the semester, the main focus was on understanding the problem and shaping the idea. This included researching existing budgeting tools, pregnancy apps, and financial planning supports to understand what already exists and where users currently struggle. From this research, we identified a clear gap: while there are tools for budgeting and separate tools for pregnancy tracking, there is no integrated platform that links financial planning directly to pregnancy milestones. During this period, we also worked on refining the core concept, identifying key features, discussing feasibility, and defining what would realistically be achievable within the timeframe of the project.

Once the idea was agreed and approved, the next stage focused on technical planning and preparation. We researched suitable technologies and confirmed that Django would be used for backend development, MySQL for structured and secure data storage, and Chart.js for displaying financial information visually. During this time, we also designed our initial system architecture, decided on a three-tier approach (frontend, backend, database), and started planning how data and user interactions would flow through the system. The project repository was created, the folder structure was set up, and we began drafting early interface concepts for key screens such as the dashboard, login, and budgeting areas.

System design and prototyping followed. This involved formalising the structure of the platform, defining models for users, budgets, expenses, milestones, and forecasts, and planning the relationships between them. Wireframes and early design sketches were created to help visualise how users would navigate the system. During this stage, the focus was on clarity, simplicity, and ensuring the system would be intuitive for expecting parents.

The next major phase is backend development. This phase includes setting up the Django project properly, implementing secure user authentication, and building the database models to store user and financial information. Key business logic will be implemented here, such as handling budgeting calculations, maternity leave adjustments, and linking financial data to pregnancy stages. Initial

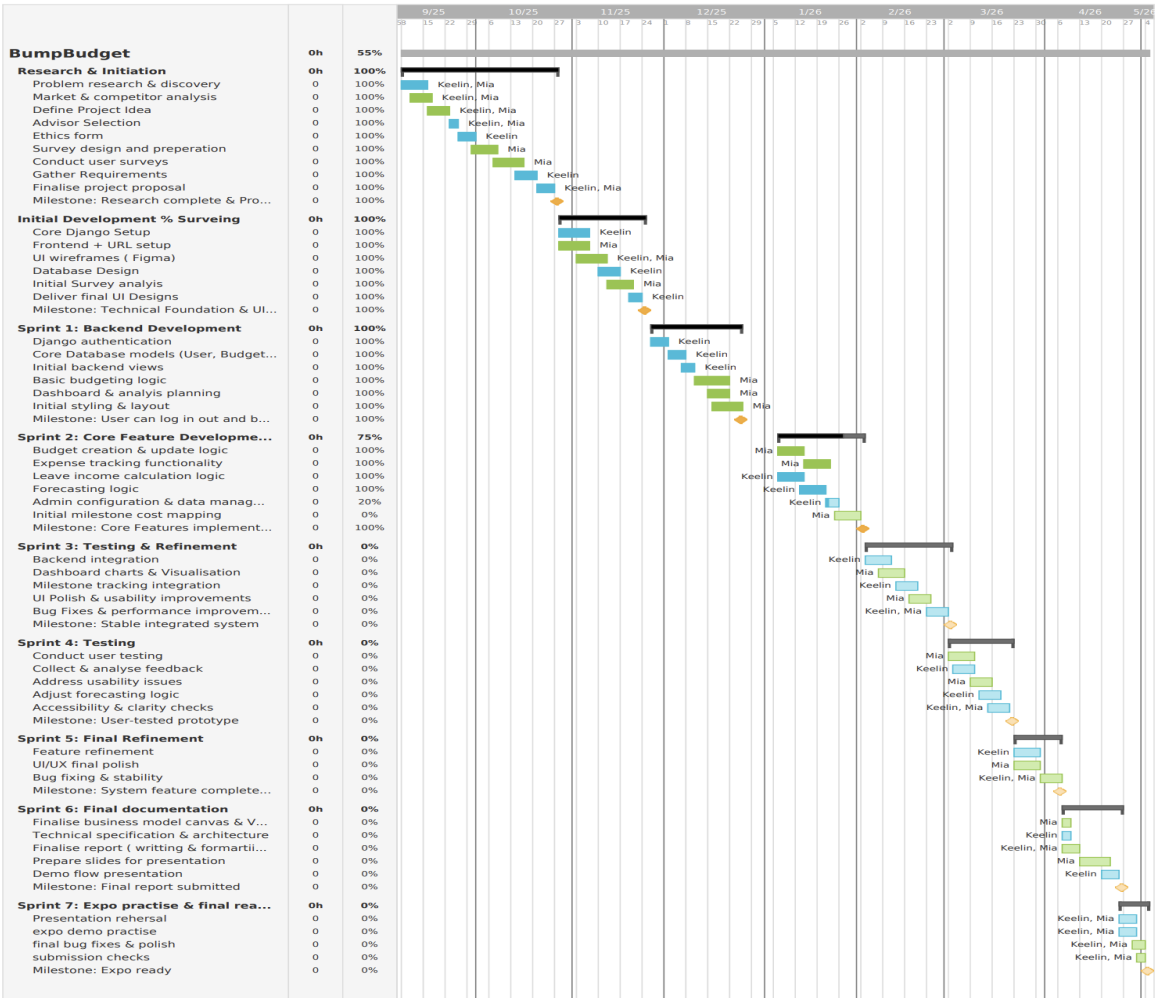
forecasting functionality will also be introduced, starting with structured rule-based predictions and potentially expanding further depending on time.

Frontend development will run alongside backend work. This involves building the actual user interface using Django templates, CSS, and JavaScript. The goal during this stage is to make the system visually clear, user-friendly, and responsive across devices. Chart.js will be integrated to transform financial data into interactive charts and graphs, helping users easily understand their spending and forecasts.

Towards the later part of the project, the focus will move to integration, testing, and improvement. This will include testing the full system to ensure it functions correctly, is secure, and provides a smooth user experience. Any issues found will be fixed, and features will be refined based on feedback. Finally, the project will move into the documentation and finalisation phase, where we will prepare the written report, complete all required deliverables, and ensure the system is ready for submission and demonstration.

### 6.3 Gantt Chart *(Fig 4)*

The BumpBudget project timeline outlines the development phases from initiation to final delivery, ensuring a structured and realistic workflow. Early stages focus on research and planning, followed by backend and frontend implementation using Django, MySQL, and Chart.js. Testing, refinement, and documentation ensure the final product meets user needs and technical standards. This schedule provides a clear visual guide for tracking progress and coordinating team responsibilities throughout the 2025–2026 academic year.



*(Fig 4) - This Gantt chart outlines the project timeline, showing the planned phases, key tasks, and milestones for completing the machine learning assessment, from dataset preparation and model development to evaluation and final reporting.*

## 7. References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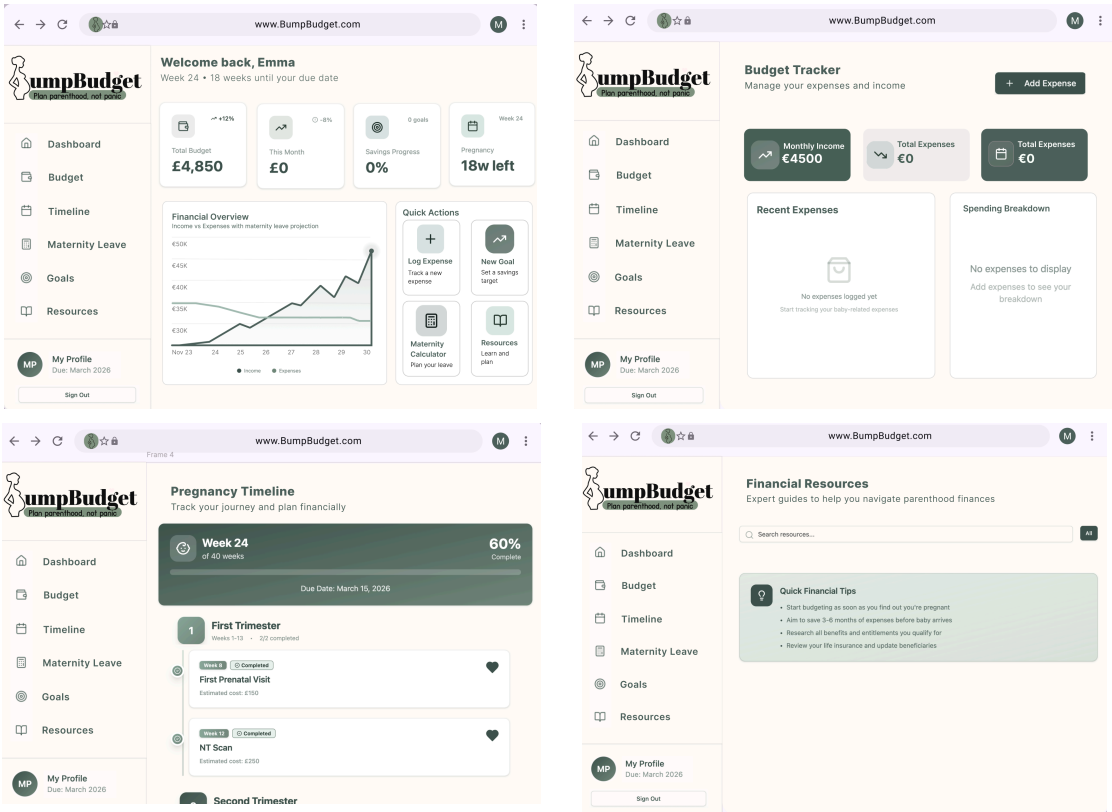
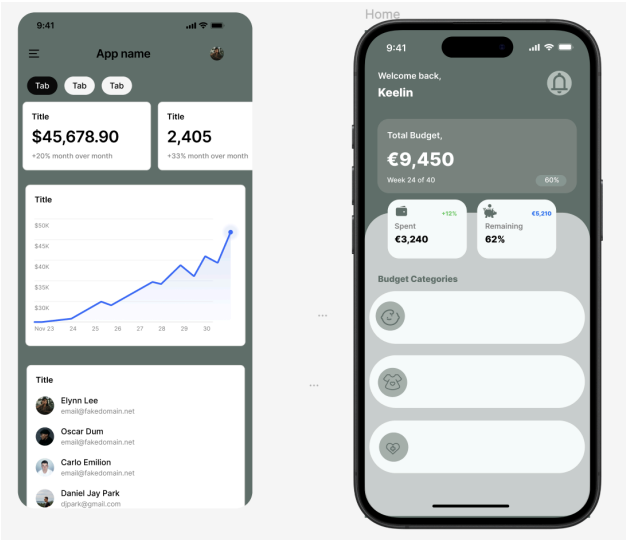
# 8. Appendices

### *Website Prototype (Fig 5)*



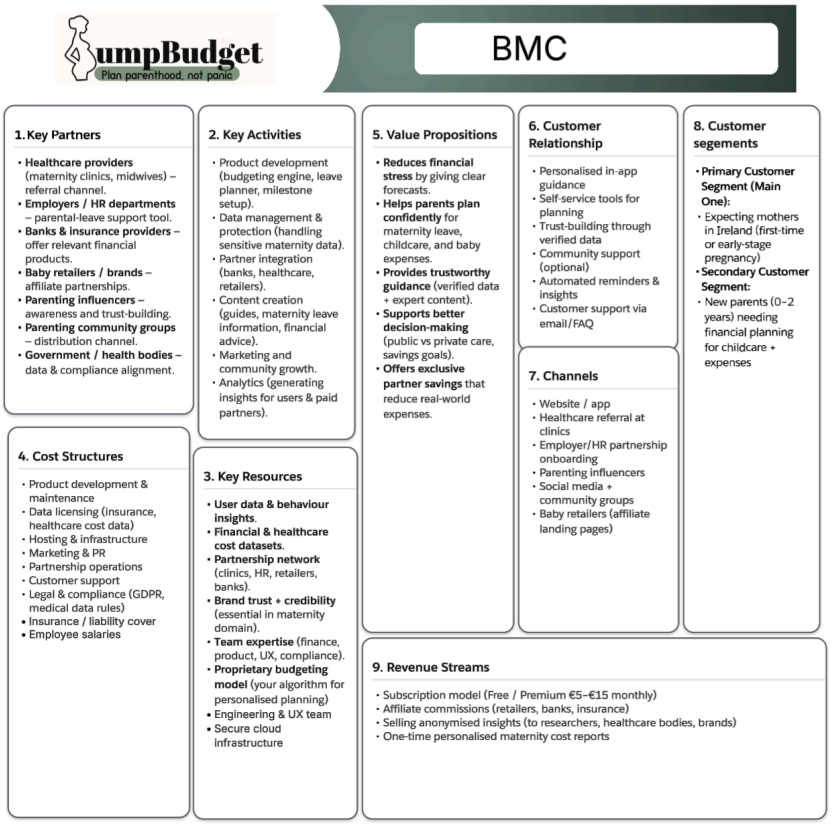

*(Fig 5) presents the high-fidelity web prototype of BumpBudget, illustrating the main user interfaces including the dashboard, budget tracking, pregnancy timeline, and educational resources.*

### *Mobile Prototype (Fig 6)*



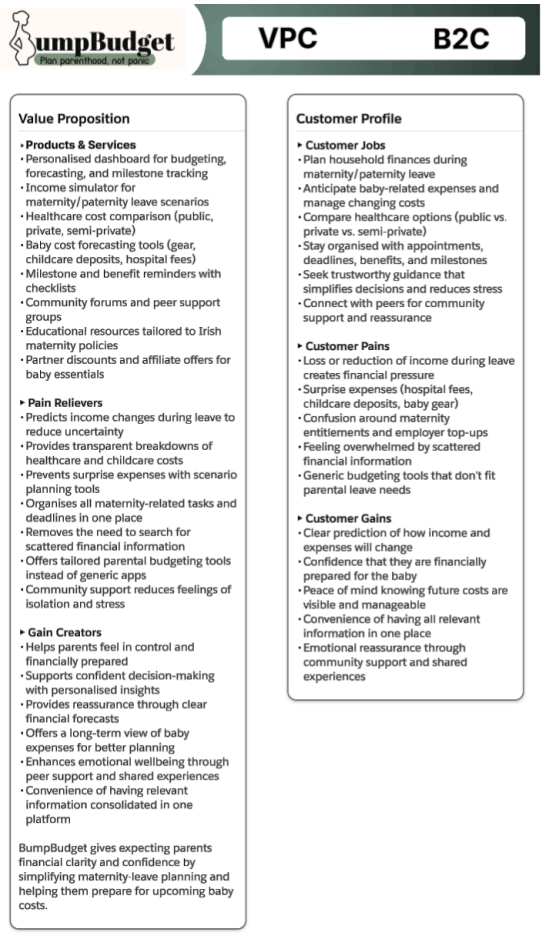

*(Fig 6) shows the mobile prototype of BumpBudget, demonstrating a mobile-first design approach and how key budgeting and forecasting features are presented on smaller devices.*

## *Business Model Canvas (Fig 7)*

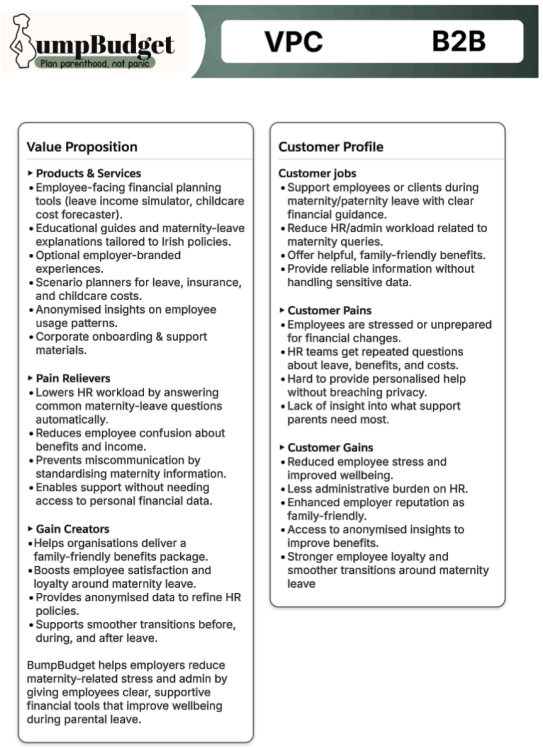



*(Fig 7) illustrates the Business Model Canvas for BumpBudget, outlining key partners, activities, value propositions, customer segments, and revenue streams considered during project planning.*

## *Value Proposition Canvas B2B (Fig 8)*




## *Value Proposition Canvas B2C (Fig 9)*



*(Fig 8) presents the B2B Value Proposition Canvas, highlighting how BumpBudget addresses organisational needs related to maternity leave planning, employee wellbeing, and cost transparency.*

*(Fig 9) presents the B2C Value Proposition Canvas, demonstrating how BumpBudget supports expecting parents through personalised financial forecasting, budgeting tools, and stress reduction.*

***Survey Link***

*A user survey was conducted to gather initial feedback and validate user needs; the survey link is provided below for reference.*

***Form Link***